TECHNICAL PROJECT REPORT

Long-Context Retrieval Fidelity Engine

A framework-free evaluation of chunking, dense retrieval, BM25, and Reciprocal Rank Fusion

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Python 3.11 | NumPy | Extractive retrieval

ABSTRACT

This project builds and evaluates a transparent retrieval pipeline for long, structured SEC Form 10-K filings. It compares fixed-window and sentence-aware chunking, independently encoded MiniLM embeddings with custom NumPy cosine search, BM25 lexical retrieval, and Reciprocal Rank Fusion (RRF). A 30-query benchmark measures evidence retrieval at beginning, middle, and end positions. Results show that boundary-aware chunking performs best among the tested configurations and RRF improves ranking metrics over dense retrieval, while the small, position-stratified benchmark does not establish a causal “lost-in-the-middle” effect in a generative LLM.

10	555,426	3,957	30
SEC FILINGS	SOURCE WORDS	SELECTED CHUNKS	BENCHMARK QUERIES

KEY FINDINGS

01 **Sentence-aware chunking.** Selected index: 3,957 chunks, 81.53% boundary cleanliness, Hit@5 0.4333, MRR@10 0.2984. Fixed-192 and fixed-96 baselines score lower on Hit@5 and MRR@10.

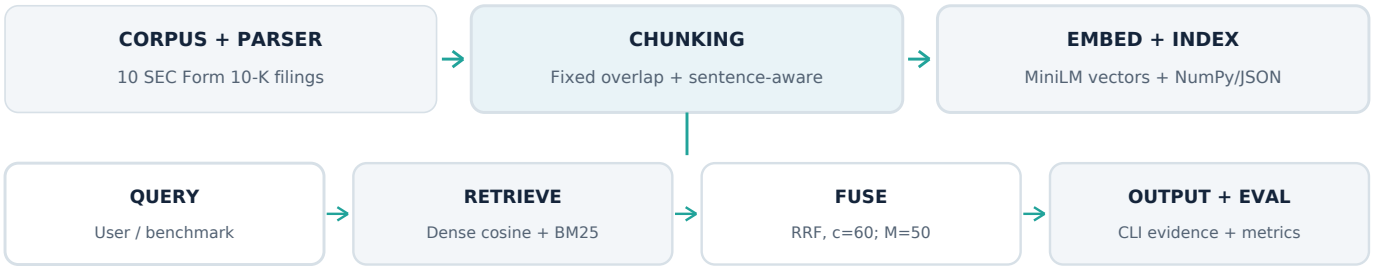
02 **Hybrid retrieval improves rank quality.** RRF reaches MRR@10 0.3823 versus 0.2984 for Dense and 0.3708 for BM25; Hit@10 is 0.5667 versus 0.5000 for each baseline.

03 **Recall@5 is not improved over Dense.** RRF Recall@5 = 0.4333, equal to Dense, and above BM25 at 0.4000. RRF increases MRR@5 from 0.2894 to 0.3678 versus Dense.

04 **Position results are confounded.** The 30 queries are split 10/10/10 by source position. Table/numeric evidence has lower Dense MRR@10 than narrative evidence; content and query selection prevent causal position claims.

Interpretation guardrail. This pipeline embeds each chunk independently. Position-stratified retrieval scores do not directly test the classic full-prompt self-attention effect described by “Lost in the Middle.”

System design and evaluation method



CORPUS AND BENCHMARK

The corpus contains ten SEC Form 10-K filings from technology companies, totaling 555,426 whitespace-delimited words; each document exceeds 5,000 words. The benchmark uses 30 evidence-grounded queries, three per company, with ten targets in each evaluated source-position band: beginning <10%, middle 40-60%, and end >=90%. Position is annotated from a word-based source offset; the CLI separately reports a character-midpoint depth.

CHUNKING TRADE-OFF

Configuration	Chunks	Mean tokens	Boundary cleanliness	Hit@5	MRR@10
Fixed window, 96	9,198	95.94	13.67%	0.2333	0.1946
Fixed window, 192	4,599	191.84	13.35%	0.3333	0.2296
Sentence-aware, 192 target	3,957	171.90	81.53%	0.4333	0.2984

Boundary cleanliness is the proportion of chunks ending on an accepted sentence boundary or table-row delimiter; the historical JSON field name “semantic_completeness” is not a general semantic-quality metric. The sentence-aware configuration was selected using this same 30-query benchmark, so the reported comparison may contain selection bias.

RETRIEVAL METHODS

DENSE COSINE

`all-MiniLM-L6-v2` creates 384-dimensional vectors. For unit vectors, cosine similarity equals the dot product: **cos(q,d) = q · d**. Ranking scans the NumPy embedding matrix.

BM25 LEXICAL

score(D,Q) = Σ IDF(q) · f(q,D)(k1+1) / [f(q,D) + k1(1-b+b|D|/avgdl)]. Parameters: **k1 = 1.5**, **b = 0.75**. Term frequency and IDF favor distinctive names, dates, and statutory terms.

RRF FUSION

Combines top-50 Dense and BM25 rankings: **RRF(d) = Σ 1/(c + rank(d))**, with **c = 60** and 1-based ranks. A rank-5 result from both scores 2/65 = 0.03077.

Representation settings: model inference length configured at 256 tokens; selected sentence-aware chunks have mean 171.90 tokens (min 25, median 177, max 192), with 3,815 of 3,957 chunks exceeding 128 tokens. The model card reports 128-token fine-tuning sequences; any specific late-token attenuation effect remains an untested hypothesis.

Evaluation relevance rule. A result is relevant when it belongs to the correct source document and encloses the annotated character span or contains the annotated evidence text. Recall is macro-averaged across queries; a query with no eligible relevant chunks is assigned recall 0.0.

EVALUATION METRICS

Hit@K = fraction of queries with at least one relevant chunk in the top K.
Precision@K = (1/N) Σ $|R_i,K|$ / K
Hit@K and precision are macro-averaged over the query set.

Recall@K = (1/N) Σ $|R_i,K|$ / T_i; if T_i = 0, $recall_i$ = 0.
MRR@K = (1/N) Σ reciprocal rank of the first relevant result when rank ≤ K, otherwise 0.

ASSIGNMENT COVERAGE SNAPSHOT

The implementation covers corpus ingestion (10 long documents), two chunking strategies, real embeddings, custom cosine and top-K search, BM25, a 30-query position-stratified benchmark, a baseline chart, RRF re-evaluation, a CLI with debug output, and before/after result artifacts. Fresh-environment setup was measured at about 12 minutes end-to-end, so the original under-five-minute setup target was not met in the reported clean-clone run.

Overall retrieval results

All three strategies were evaluated on the same 3,957-chunk sentence-aware index and 30-query benchmark. Values below are from the supplied precision/recall comparison artifact; metrics are macro-averaged across queries where applicable.

Mode	Hit@1	Hit@5	Hit@10	Recall@5	MRR@5	MRR@10
Dense	0.2333	0.4333	0.5000	0.4333	0.2894	0.2984
BM25	0.3333	0.4000	0.5000	0.4000	0.3556	0.3708
RRF	0.3333	0.4333	0.5667	0.4333	0.3678	0.3823

Main outcome. RRF improves MRR@10 by +0.0839 over Dense (+28.1% relative) and improves Hit@10 from 0.5000 to 0.5667. It does not improve Recall@5 over Dense (both 0.4333), though it exceeds BM25 by +0.0333 at K=5.

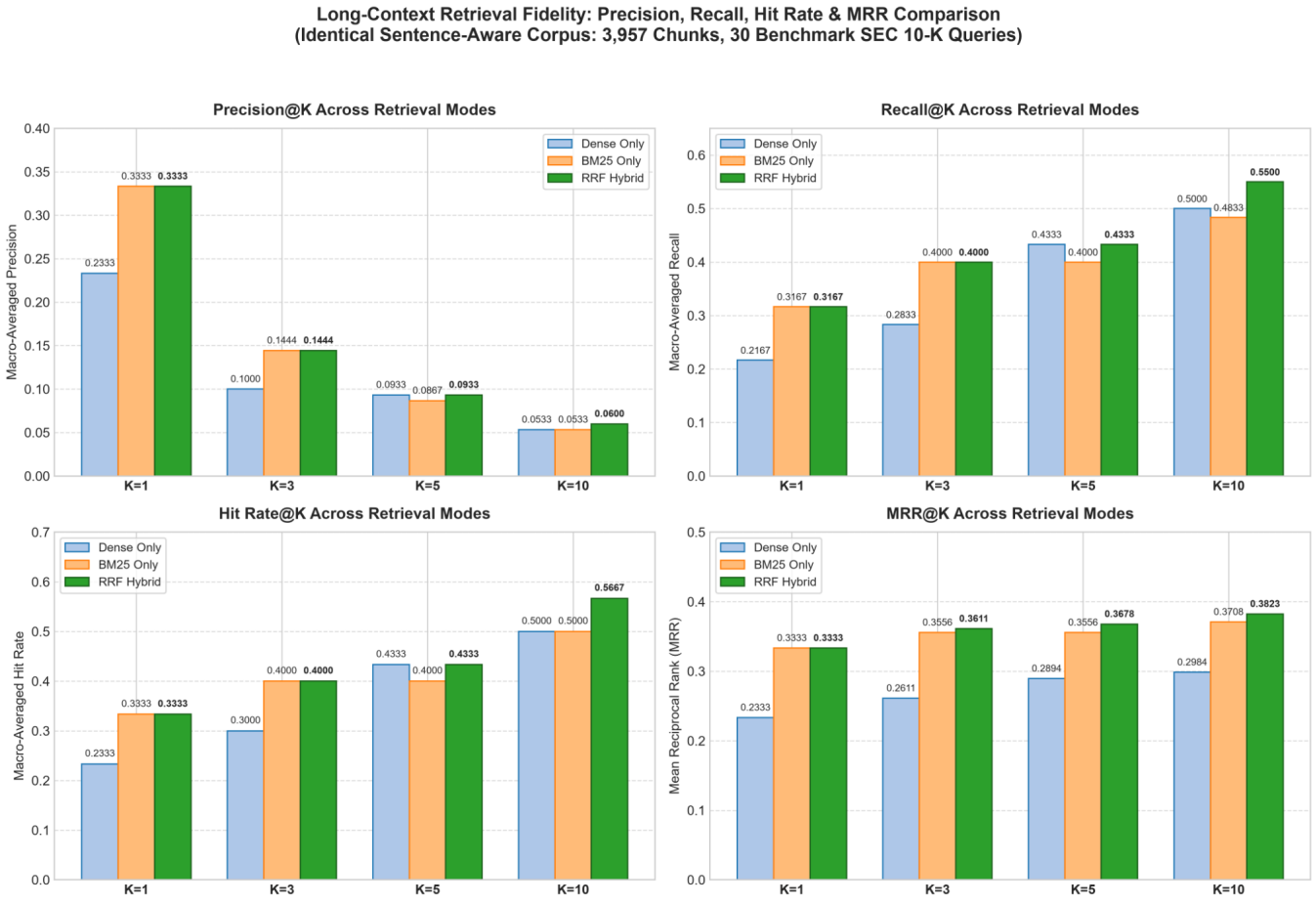


Figure 1. Precision, macro-averaged recall, hit rate, and MRR across K = 1, 3, 5, 10.

READING THE K TRADE-OFF

At K=1, BM25 and RRF each hit 10 of 30 queries (Hit@1 = 0.3333), compared with 7 for Dense (0.2333). At K=10, RRF reaches 0.5667 Hit@10 while Dense and BM25 each reach 0.5000. Precision naturally falls as K grows because the query annotations typically identify only one or two relevant chunks. Equal aggregate metrics at K=5 do not mean that methods returned identical chunks for each query.

Chunking takeaway. Compared with fixed-192, sentence-aware chunking uses 642 fewer chunks (-14.0%) while increasing Hit@5 by 0.1000 and MRR@10 by 0.0688 on the evaluated benchmark. Because the same queries were used to choose the configuration, this is an observed comparison rather than an independent held-out estimate.

Position, structure, and mitigation

Position-bucket values are descriptive and exploratory (n=10 per bucket). The main sentence-aware comparison shows Dense MRR@10 of 0.2676 / 0.1650 / 0.4625 for beginning / middle / end, compared with RRF at 0.3600 / 0.2644 / 0.5225. The separate historical fixed-192 chart is labeled explicitly below.

Position (n=10)	Dense MRR@10	BM25 MRR@10	RRF MRR@10	Table queries
Beginning	0.2676	0.4333	0.3600	2 / 10
Middle	0.1650	0.2458	0.2644	4 / 10
End	0.4625	0.4333	0.5225	4 / 10

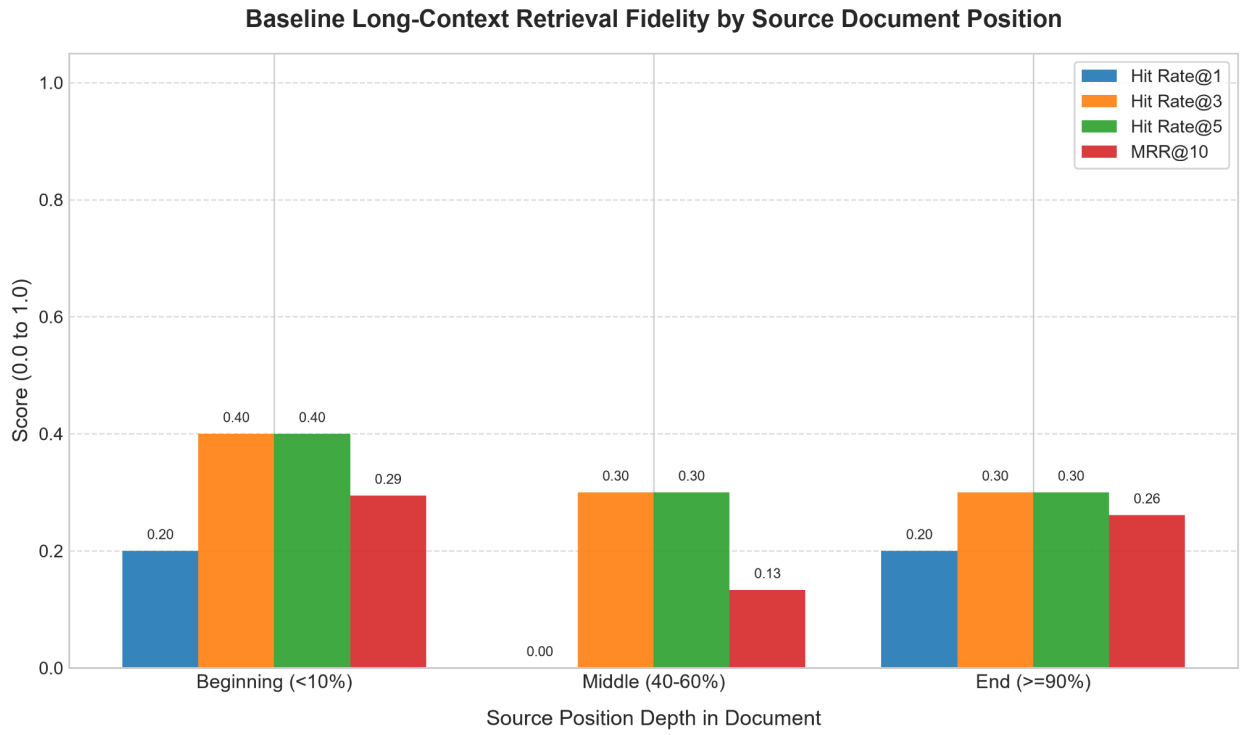


Figure 2. Historical fixed-192 dense baseline. This is the chunking-comparison baseline, not the selected sentence-aware index.

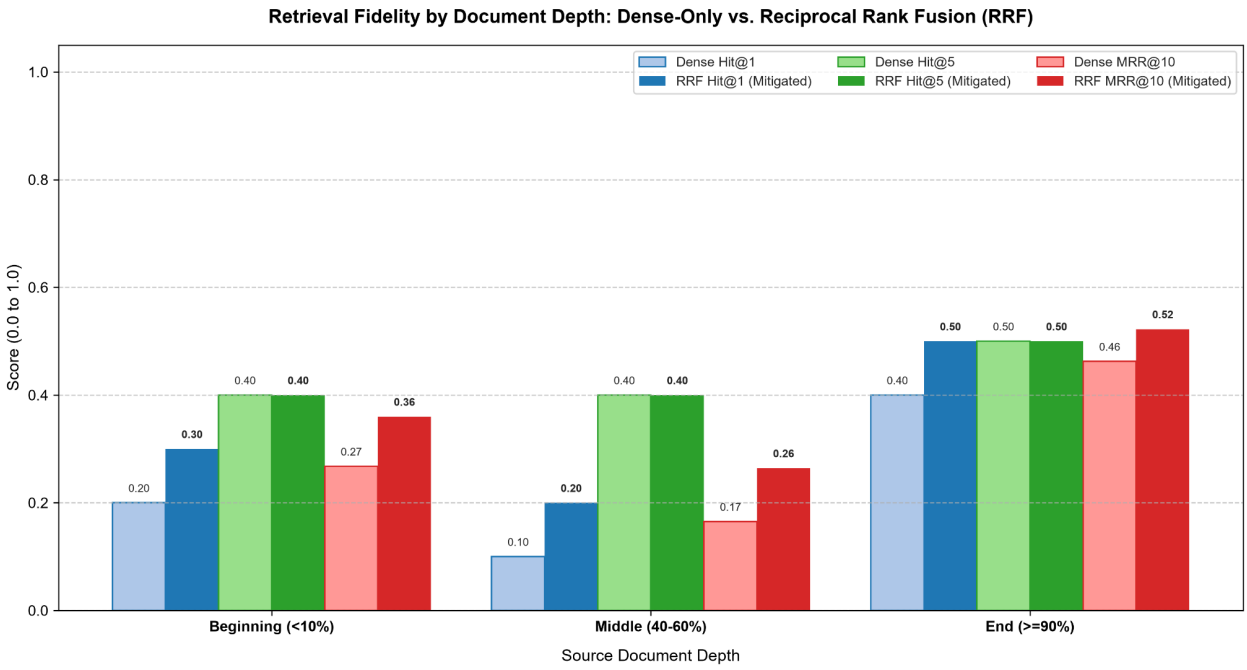


Figure 3. Dense-only versus RRF on the sentence-aware index by document depth.

Content-type analysis classifies 10 queries as table/numeric and 20 as narrative. Dense MRR@10 is 0.1893 for table/numeric evidence versus 0.3529 for narrative. The middle and end buckets each contain four table queries; table share alone does not explain the full position gap. Within narrative evidence, middle Dense MRR@10 (0.2333, n=6) is not below beginning (0.1917, n=8).

Validity, limitations, and reproducibility

STATISTICAL INTERPRETATION

The supplied validity analysis uses 10,000 paired bootstrap resamples (seed 42; 95% percentile intervals), exact two-sided McNemar tests for paired Hit@K outcomes, and an exact sign test for paired MRR@10 differences with zero-difference pairs excluded. The 15 headline tests cover three retriever pairs across Hit@1, Hit@3, Hit@5, Hit@10 and MRR@10; the Bonferroni threshold is 0.00333. No comparison crosses that corrected threshold. The sample is exploratory, not a powered demonstration of equivalence.

Paired comparison	MRR@10 delta	Bootstrap 95% CI (unadjusted)	Exact sign-test p
RRF - Dense	+0.0839	[+0.0169, +0.1676]	0.1094
RRF - BM25	+0.0115	[-0.0820, +0.1046]	1.0000
BM25 - Dense	+0.0725	[-0.0456, +0.1975]	0.2266

LIMITATIONS AND HONEST INTERPRETATION

- Small sample: 30 queries total and only 10 per position bucket; one query shifts a bucket hit rate by 10 percentage points.
- Query wording was developed from target evidence, which can favor lexical matching. Exact-span/sub-string relevance can penalize alternative valid passages.
- Chunking configuration was selected on the same benchmark used for evaluation; independent held-out validation is needed.
- The benchmark tests retrieval of independently encoded chunks, not an LLM consuming a long prompt. Position differences can reflect tables, section vocabulary, query difficulty, and boundaries.
- The PDF does not establish statistical equivalence or production readiness; it reports the observed results and the available evidence only.

REPRODUCTION AND PROJECT ACCESS

Sample query (from the repository root):

```
python -m src.cli --question "What were Amazon's consolidated net sales in 2025?" --mode rrf --top-k 5
python -m pytest -q
```

The project provides a precomputed 3,957-chunk index and three result charts. The reported clean-clone setup (clone, environment, CPU PyTorch, dependencies, query and tests) took about 12 minutes; this does **not** meet the original under-five-minute fresh-setup target, although the cached-index query is reusable after installation. A prior project run reported 82 passing tests; this report did not independently rerun the repository.

Repository: https://github.com/AyushmanGupta21/long-context-retrieval-fidelity

CONCLUSION AND NEXT STEPS

The tested evidence supports a practical result: sentence-aware boundaries outperformed the two fixed-window configurations on this benchmark, and RRF improved MRR@10 and Hit@10 over Dense alone. It did not improve Recall@5 over Dense, and the 30-query evaluation cannot isolate a causal document-position effect.

01 VALIDATE

Expand to 150+ held-out questions; have a second reviewer adjudicate evidence spans.

02 STRUCTURE

Preserve table headers and units across chunks; measure table retrieval separately.

03 HARDEN

Evaluate cross-encoder reranking; pin dependency versions and add CI checks.

Selected references: Liu et al., “Lost in the Middle: How Language Models Use Long Contexts” (arXiv:2307.03172, 2023); Cormack, Clarke & Büttcher, “Reciprocal Rank Fusion Outperforms Condorcet and Individual Rank Learning Methods” (SIGIR 2009); Robertson & Zaragoza, “The Probabilistic Relevance Framework: BM25 and Beyond” (FnT Information Retrieval, 2009); Reimers & Gurevych, “Sentence-BERT” (EMNLP-IJCNLP 2019); `sentence-transformers/all-MiniLM-L6-v2` model card; SEC EDGAR filings.